Project Proposal: Public Transport Route Planning and Simulation

Objective

There are two main domains:

- The Urban Transport domain, which includes stops, lines, vehicles, timetables, and passenger demand.
- The Geographical Mapping domain, which represents city areas, roads, zones, and distances between stops.

The problem to address is: Given a city map and potential stops, how can we automatically plan public transport routes that ensure high coverage while minimizing total travel time and operational cost?
The goal is to design a model-based framework that transforms city and transport data into optimized route models, and simulates performance like coverage, waiting time and transport efficiency.

Models

I anticipate the use of 4 metamodels (see Figure 1):
1. TransportMM: Represents stops, lines, vehicles, schedules, and passengers.
2. CityMapMM: Represents the physical layout, zones, roads, and distances.
3. MathsMM: Defines optimization inputs, graphs, constraints, and objectives.
4. ResultMM: Captures simulation outcomes coverage rate, travel time, and utilization.

There are three domains: Urban Transport, Geographical Mapping, and Optimization/Simulation. TransportMM and CityMapMM combine through a M2M to form MathsMM. An external tool consumes this model and produces results, which are converted into ResultMM via T2M and then reflected back using M2M.

Tool integration external tools that produce or consume models

An external optimization/simulation tool will be integrated:
- For optimization: MATLAB or Python, with libraries to compute optimal routes and other computations needed.
- For simulation: Eclipse SUMO (Simulation of Urban MObility) is a good choice to evaluate this project since it's an open source traffic simulation [1].

The M2T transformation generates solver inputs, and T2M parses outputs back into the modeling environment.

How will the model be created/updated?
- The CityMap model will be manually created and updated.
- The Transport model defines lines and demand manually.
- The Maths model will be automatically generated via M2M.

- The Result model will be created automatically through T2M.
- The Transport model will be updated automatically through M2M with optimized routes.

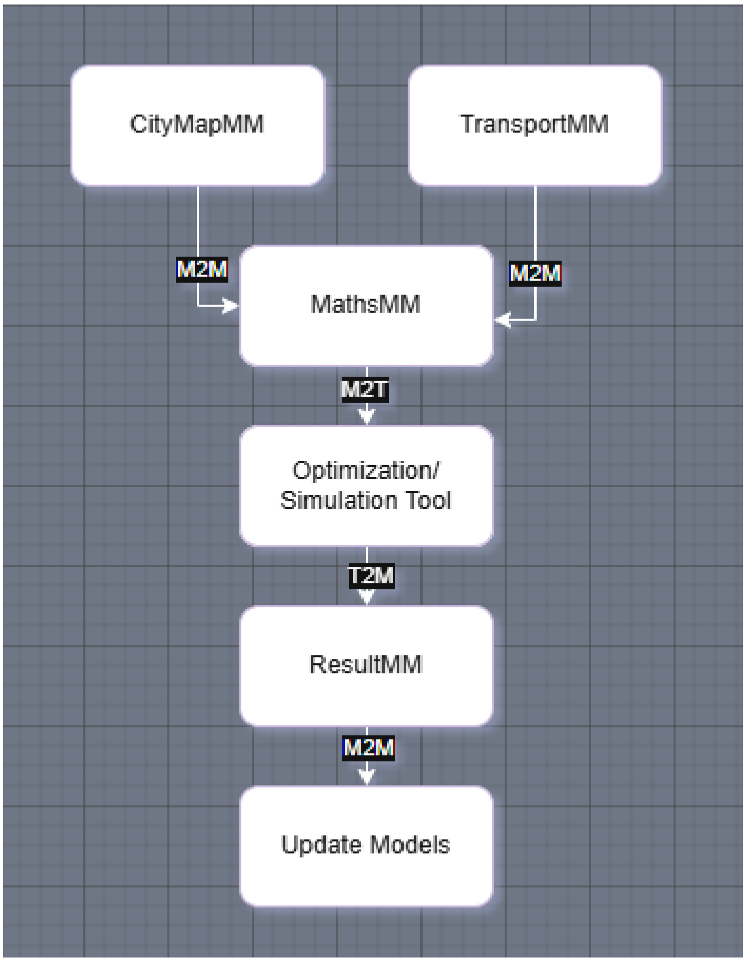

Figure 1 - *Concept flow*

Transformations

1. M2M (CityMap + Transport → Maths): Combine city layout and operational data into a math model.
2. M2T (Maths → Solver Input): Generate solver or simulation input files.
3. T2M (Tool Output → Result): Convert simulation results into structured data.
4. M2M (Result → Transport): Update the transport model with new routes.
5. M2T (Result → Visualization): Produce visual maps and performance reports.

The M2M and M2T transformations execute sequentially to prepare solver input. After the simulation, T2M and M2M execute to get results and update models. The visualization M2T runs to generate maps and other data.

References

[1] Eclipse Foundation, “SUMO — Simulation of Urban Mobility,” Eclipse, 2025. [Online]. Available: https://www.eclipse.org/sumo/ [Accessed: Oct. 13, 2025].